

# Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: ASAP++ | Context: Qatari and Arab High School Arabic Essay Writers (Grades 10–12)

**Input Ontology definition:** Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The benchmark's category inventory consists of three essay types — argumentative/persuasive, source-dependent, and narrative/descriptive [\[Q12, Q13, Q14, Q15\]](#) — drawn from a US middle-school context. The deployment requires coverage of literary analysis, religious-text commentary, and cultural commentary genres that are standard in Arab high school curricula but entirely absent from the benchmark. The Qatar MOEHE curriculum confirms Arabic Literature as a distinct subject and Islamic Education as compulsory [\[WEB-11, WEB-5\]](#), yet no AES benchmark covers these genres in Arabic. The expert elicitation flagged IO as HIGH priority precisely because the benchmark's taxonomy omits the test-case types the deployment requires. Even within the persuasive genre that nominally overlaps, scoring logic is type-specific and embedded in English pedagogical norms [\[Q43, Q44, Q45\]](#).

| ID                     | Evidence                                                                                                                                                                                                                                                                                         |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Q12</a>    | "There are 3 types of essays in the dataset." (p.2)                                                                                                                                                                                                                                              |
| <a href="#">Q13</a>    | "Argumentative / Persuasive essays - These are essays where the prompt is one in which the writer has to convince the reader about their stance for or against a topic (for example, free speech in public colleges)." (p.2)                                                                     |
| <a href="#">Q14</a>    | "Source-dependent responses - These essays are responses to a source text, where the writer responds to a question about the text (for instance, describing the writer's opinion about an incident that happened to him in the text)." (p.2)                                                     |
| <a href="#">Q15</a>    | "Narrative / Descriptive essays - These are essays where the prompt requires us to describe / narrate a story." (p.2)                                                                                                                                                                            |
| <a href="#">Q43</a>    | "For source-dependent essays, we found out that the most important feature for content was length, while for argumentative / persuasive essays, it was coherence and cohesion features, followed by length." (p.4)                                                                               |
| <a href="#">Q44</a>    | "For source-dependent essays, the coherence and cohesion feature set is the most important feature set for each of the other 3 attributes." (p.4)                                                                                                                                                |
| <a href="#">Q45</a>    | "For persuasive / argumentative essays, coherence and cohesion features are the most important features for 4 of the 5 attributes." (p.4)                                                                                                                                                        |
| <a href="#">WEB-5</a>  | Islamic Education is compulsory in Qatar national curriculum schools for Qatari pupils and Arab passport holders ( <a href="https://www.qisweb.qis.org/arabic-department/">https://www.qisweb.qis.org/arabic-department/</a> )                                                                   |
| <a href="#">WEB-11</a> | IMPACT-se review of Qatar 2025–2026 curriculum confirms Arabic Literature and politically/religiously contextualized content ( <a href="https://www.impact-se.org/wp-content/uploads/Qatar-Report-2026-final.pdf">https://www.impact-se.org/wp-content/uploads/Qatar-Report-2026-final.pdf</a> ) |

## Strengths

- The conceptual move toward attribute-level (multi-trait) scoring across multiple essay types [\[Q4, Q18\]](#) is methodologically analogous to what an Arabic formative feedback system would target, as later Arabic AES work like LAILA's seven-trait rubric demonstrates [\[WEB-12\]](#).

| ID                     | Evidence                                                                                                                                                                                                     |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Q4</a>     | "Our contribution is the scoring of individual attributes of the essays, like content, organization, style, etc. in the ASAP dataset for the remainder of the essays." (p.1)                                 |
| <a href="#">Q18</a>    | "Based on the types of essays, there are 2 sets of attributes." (p.2)                                                                                                                                        |
| <a href="#">WEB-12</a> | LAILA used annotators trained on Arab high school writing — example of appropriate annotator profile for this deployment ( <a href="https://arxiv.org/abs/2512.24235">https://arxiv.org/abs/2512.24235</a> ) |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                            |
|------|----------------------------------------------------------------|
| IO-1 | Identify test case categories required for regional deployment |
| IO-2 | Check if taxonomy omits regionally relevant categories         |
| IO-3 | Check if taxonomy includes irrelevant categories               |
| IO-4 | Document category gaps harming content validity                |

### Checklist responses:

**IO-1:** The deployment requires at least five genres: persuasive/argumentative (ةيعانقإ ةلاقم), explanatory/expository, literary analysis (ةلأق ليلحت ةلاقم), religious-text commentary (ريسفت / ريدي صن لمأت), and cultural commentary (ةلاقم ةلأق), grounded in Qatar MOEHE Arabic Language and Arabic Literature curricula [WEB-10, WEB-11, WEB-5].

**IO-2:** Yes — literary analysis, religious-text commentary, and cultural commentary are all absent from the benchmark, which covers only argumentative/persuasive, source-dependent, and narrative/descriptive types [Q12, Q13, Q14, Q15].

**IO-3:** The narrative/descriptive category and source-dependent responses keyed to specific US source texts [Q14, Q15] introduce categories not aligned with formal Arab high school assessment, where literary analysis and religious-text engagement are more central [WEB-11].

**IO-4:** Category gaps — absence of literary analysis, religious commentary, and cultural commentary — represent severe construct underrepresentation. The benchmark also embeds genre-specific scoring rules (length-weighted for source-dependent; coherence/cohesion-weighted for argumentative) [Q43, Q44, Q45] that would not transfer to Arabic rhetorical norms even within the nominally overlapping argumentative genre.

| ID     | Evidence                                                                                                                                                                                                                                                                                         |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| Q14    | "Source-dependent responses - These essays are responses to a source text, where the writer responds to a question about the text (for instance, describing the writer's opinion about an incident that happened to him in the text)." (p.2)                                                     |
| Q15    | "Narrative / Descriptive essays - These are essays where the prompt requires us to describe / narrate a story." (p.2)                                                                                                                                                                            |
| Q43    | "For source-dependent essays, we found out that the most important feature for content was length, while for argumentative / persuasive essays, it was coherence and cohesion features, followed by length." (p.4)                                                                               |
| Q44    | "For source-dependent essays, the coherence and cohesion feature set is the most important feature set for each of the other 3 attributes." (p.4)                                                                                                                                                |
| Q45    | "For persuasive / argumentative essays, coherence and cohesion features are the most important features for 4 of the 5 attributes." (p.4)                                                                                                                                                        |
| WEB-5  | Islamic Education is compulsory in Qatar national curriculum schools for Qatari pupils and Arab passport holders ( <a href="https://www.qisweb.qis.org/arabic-department/">https://www.qisweb.qis.org/arabic-department/</a> )                                                                   |
| WEB-10 | Qatar MOE Arabic Language curriculum framework anchors Grades 1–12 writing standards ( <a href="http://www.sec.gov.qa/En/Education/Pages/GradeAndSubject.aspx">http://www.sec.gov.qa/En/Education/Pages/GradeAndSubject.aspx</a> )                                                               |
| WEB-11 | IMPACT-se review of Qatar 2025–2026 curriculum confirms Arabic Literature and politically/religiously contextualized content ( <a href="https://www.impact-se.org/wp-content/uploads/Qatar-Report-2026-final.pdf">https://www.impact-se.org/wp-content/uploads/Qatar-Report-2026-final.pdf</a> ) |
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## Information Gaps

- Whether Quranic/religious-text commentary essays are graded within the Arabic Language curriculum versus only Islamic Education is not resolvable from open sources.

- Cross-country curricular variation among Arab nations for genre coverage and weighting is undocumented in open sources.

Requires Expert Verification

- Genre inventory and weighting in Qatar MOEHE Grades 10–12 Arabic writing assessment
- Whether literary analysis essays follow distinct rubric structures across Arab national curricula

Remediation

**Gap 1: Benchmark covers only argumentative/persuasive, source-dependent, and narrative/descriptive genres; deployment requires literary analysis, religious-text commentary, and cultural commentary.**

**Recommendation:** Construct a genre inventory aligned with Qatar MOEHE Arabic Language and Arabic Literature curricula [WEB-10, WEB-11], using LAILA's prompt set [WEB-12] as a partial proxy and commissioning new Qatari teacher-authored prompts for the three missing genres.

| ID     | Evidence                                                                                                                                                                                                                                                                                         |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WEB-10 | Qatar MOE Arabic Language curriculum framework anchors Grades 1–12 writing standards ( <a href="http://www.sec.gov.qa/En/Education/Pages/GradeAndSubject.aspx">http://www.sec.gov.qa/En/Education/Pages/GradeAndSubject.aspx</a> )                                                               |
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